Problem Scoping :-

Use cases :- Technology Adoption Prediction System.

- **Technology Companies**

→ Notice that many new technologies, apps and digital products are launched every year.
→ Some technologies are adopted very quickly while others fail to attract users.
→ Want to predict how quickly people will adopt a new technology and increase adoption rate.

- They approach an AI Team / Organization.

→ Asked to build an AI system which can predict whether a new technology will be adopted quickly or slowly by users.
→ Sounds very straight forward but the solution is not a well-defined AI Problem.

Why This Problem Matters

- Technology adoption determines the success or failure of a product.
- Low adoption can lead to financial losses for companies.
- Understanding adoption patterns helps improve product design and marketing strategies.
- It helps companies reach the right audience.

Step-1

Initial Statement

→ People are not adopting new technology fast enough.
→ Need to uncover the actual problem.

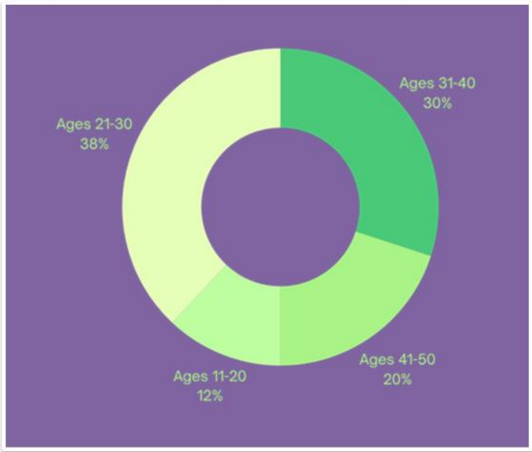
The AI team will ask themself multiple questions

Questions for stakeholders

→ **To Product Management Team**

- What is the current adoption rate?
- Since when has adoption become slow?
- Which technology products are affected?
- What factors influence adoption?

→ **Diagram: usage of technology according to age.**



→ **To Marketing Team**
- What promotion methods are currently used?
- Which user groups are targeted?
- How much user feedback is available?

→ **To Company Management**
- Why do you think this problem is important?
- What financial impact does low adoption have?
- What is the desired adoption rate?

Suppose the company reverts with following reports:

- Current adoption rate : 35%
- Desired adoption rate : 60%
- Most users stop using the technology within first month.
- Early prediction can improve marketing strategies.

➢ Step-2 : Define the problem more precisely

→ **Poor problem statement :**
- Increase technology adoption.

→ Better approach
- Predict whether users are likely to adopt a new technology within the first 30 days after launch so that the company can take early actions.

- What
  - Predict likelihood of technology adoption by users.

- When

– Within first 30 days after technology launch.

- Why

– So that companies can improve marketing, training and user engagement.

- Outcome

– Increase technology adoption rate and reduce product failure.

➢ Step-3

→ Is AI really needed?

- – Can the problem be solved without AI?
- – If only a few simple factors decide adoption, then simple rules can work.

Example:

- If product price is low → high adoption.
- If product is difficult to use → low adoption.

But real-world adoption depends on many factors:

- User age
- User experience
- Product features
- Marketing campaigns
- Social influence
- Reviews and feedback

AI should only be used when patterns are complex.

– Multiple factors interact.

– Manual rules are insufficient.

Therefore, **Machine Learning / AI is needed** to predict technology adoption accurately.

➢ Step-4

→ Possible Factors Affecting the Problem

Several factors may influence technology adoption:

– Personal Factors

- Age
- Education
- Income
- Occupation

– Social Factors

- Peer Influence
- Social Media Trends
- Online Reviews

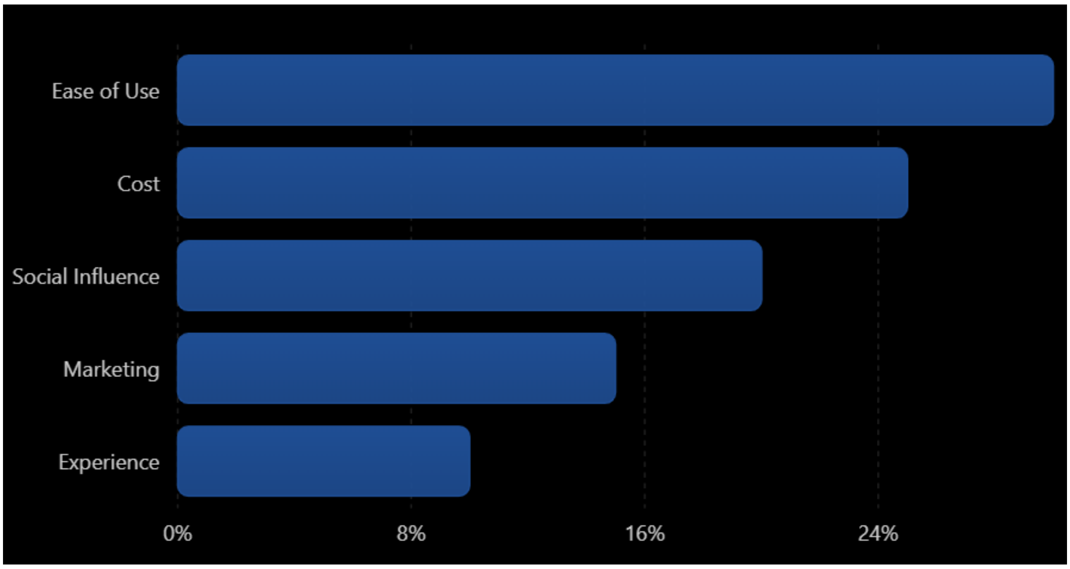
– Product Factors

- Ease of Use
- Product Features
- Reliability
- Cost

– Marketing Factors

- Advertisements
- Discounts
- Promotional Campaigns

○ **Factors influencing technology adoption**



➢ Step-5

→ Identifying Stakeholders

– The AI team identifies all the people who are affected by the problem or can provide useful information.

→ *Stakeholders Mapping:*

– Company Management
– Product Development Team
– Marketing Team
– Customers / Users

- Customer Support Team
- Sales Team

Their Interests

→ **Company Management**
- Wants higher adoption rate.
- Wants better return on investment.

→ **Marketing Team**
- Wants to know which users are likely to adopt.
- Wants to create better campaigns.

→ **Product Team**
- Wants to improve product features.

→ **Customers**
- Want useful and easy-to-use technology.

➢ Step-6

→ Define AI Task

- The AI team now defines what exactly AI needs to do.

→ *Task Type*

- Prediction Task

→ *AI Task*

- Predict whether a user is likely to adopt a new technology.

→ Machine learning formulation
- Classification problem:
  Output : Adopt = Yes | adopt = No
- Binary Classification problem

→ *Input*

- Age
- Education
- Income
- Occupation
- Previous Technology Experience
- Product Awareness
- Marketing Exposure

→ *Output*

- Likely to Adopt
- Unlikely to Adopt
- or

- Adoption Probability (%)
- Example:
- User A → 85% Adoption Chance
- User B → 25% Adoption Chance

– Risk scoring
– How likely is adopt new technology? 0to100%

➢ Step-7

→ Defining Success Metrics

- The AI team decides how success will be measured.

→ *Technical Metrics*

- Accuracy
- Precision
- Recall
- F1 Score

→ *Business Metrics*

- Increase adoption rate
- Increase active users
- Reduce product failure rate

→ *Success Criteria*

- Model accuracy above 85%
- Adoption rate improved by at least 20%
- Better targeting of marketing campaigns
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➢ Step-8

→ Defining Constraints

- The AI team identifies limitations and restrictions.

→ *Possible Constraints*

- Data Constraints
  - Limited historical data available.
  - Missing user information.
- Time Constraints
  - Prediction needed before product launch.
- Budget Constraints
  - Limited resources for data collection.
- Technical Constraints
  - Large number of users.

- Different user behaviour.

- Privacy Constraints
  - User data must remain secure.
- Ethical constraints:
  - User privacy must be protected.
  - Personal data should not be shared without permission.
  - The AI system should not discriminate based on age, gender, income, or education.
  - Predictions should be fair for all users.

➢ Step-9

→ Assumptions and Risks

- *Assumptions*

- Users provide correct information.
- Historical data represents future behaviour.
- User preferences remain relatively stable.
- Marketing data is accurate.

- *Risks*

- Incomplete data may reduce accuracy.
- User behaviour may change unexpectedly.
- New technologies may behave differently from past technologies.
- Model predictions may become outdated over time.

- *Impact*

- Wrong predictions may lead to:
- Poor marketing decisions.
- Lower adoption rate.
- Financial losses.
- Therefore, the model should be updated regularly with new data.